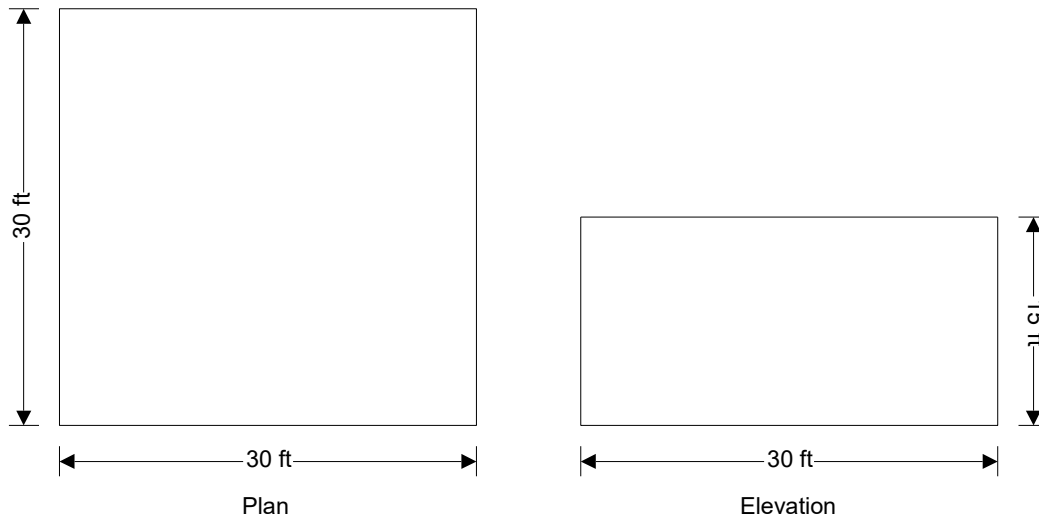


WIND LOADING

In accordance with ASCE7-16

Using the directional design method

Tedds calculation version 2.1.19



Building data

Type of roof	Flat
Length of building	b = 30.00 ft
Width of building	d = 30.00 ft
Height to eaves	H = 15.00 ft
Mean height	h = 15.00 ft

General wind load requirements

Basic wind speed	V = 120.0 mph
Risk category	II
Wind directionality factor (Table 26.6-1)	$K_d = 0.85$
Ground elevation above sea level	$Z_{gl} = 0$ ft
Ground elevation factor	$K_e = \exp(-0.0000362 \times Z_{gl}/1\text{ft}) = 1.00$
Exposure category (cl 26.7.3)	C
Enclosure classification	Enclosed buildings
Internal pressure coef +ve (Table 26.13-1)	$GC_{pi,p} = 0.18$
Internal pressure coef -ve (Table 26.13-1)	$GC_{pi,n} = -0.18$
Gust effect factor	$G_f = 0.85$
Minimum roof design wind load (cl.27.1.5)	$p_{min,r} = 8$ psf
Minimum wall design wind load (cl.27.1.5)	$p_{min,w} = 16$ psf

Topography

Topography factor not significant	$K_{zt} = 1.0$
Velocity pressure equation	$q = 0.00256 \times K_z \times K_{zt} \times K_d \times K_e \times V^2 \times 1\text{psf}/\text{mph}^2$

Velocity pressures table

z (ft)	K_z (Table 26.10-1)	q_z (psf)
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Project				Job Ref.	
Section				Sheet no./rev. 2	
Calc. by R	Date 4/26/2026	Chk'd by	Date	App'd by	Date

z (ft)	K _z (Table 26.10-1)	q _z (psf)
15.00	0.85	26.63

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.) $q_i = 26.63$ psf

Pressures and forces

Net pressure $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$

Net force $F_w = p \times A_{ref}$

Roof load case 1 - Wind 0, GC_{pi} 0.18, -C_{pe}

Zone	Ref. height (ft)	Ext pressure coefficient C _{pe}	Peak velocity pressure q _p (psf)	Net pressure p (psf)	Area A _{ref} (ft ²)	Net force F _w (kips)
A (-ve)	15.00	-0.90	26.63	-25.17	225.00	-5.66
B (-ve)	15.00	-0.90	26.63	-25.17	225.00	-5.66
C (-ve)	15.00	-0.50	26.63	-16.11	450.00	-7.25

Total vertical net force $F_{w,v} = -18.58$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 1 - Wind 0, GC_{pi} 0.18, -C_{pe}

Zone	Ref. height (ft)	Ext pressure coefficient C _{pe}	Peak velocity pressure q _p (psf)	Net pressure p (psf)	Area A _{ref} (ft ²)	Net force F _w (kips)
A	15.00	0.80	26.63	13.32	450.00	5.99
B	15.00	-0.50	26.63	-16.11	450.00	-7.25
C	15.00	-0.70	26.63	-20.64	450.00	-9.29
D	15.00	-0.70	26.63	-20.64	450.00	-9.29

Overall loading

Projected vertical plan area of wall $A_{vert_w_0} = b \times H = 450.00$ ft²

Projected vertical area of roof $A_{vert_r_0} = 0.00$ ft²

Minimum overall horizontal loading $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 7.20$ kips

Leeward net force $F_l = F_{w,wB} = -7.3$ kips

Windward net force $F_w = F_{w,wA} = 6.0$ kips

Overall horizontal loading $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 13.2$ kips

Roof load case 2 - Wind 0, GC_{pi} -0.18, +C_{pe}

Zone	Ref. height (ft)	Ext pressure coefficient C _{pe}	Peak velocity pressure q _p (psf)	Net pressure p (psf)	Area A _{ref} (ft ²)	Net force F _w (kips)
A (+ve)	15.00	-0.18	26.63	0.72	225.00	0.16
B (+ve)	15.00	-0.18	26.63	0.72	225.00	0.16
C (+ve)	15.00	-0.18	26.63	0.72	450.00	0.32

Total vertical net force $F_{w,v} = 0.65$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 2 - Wind 0, GC_{pi} -0.18, $+C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A	15.00	0.80	26.63	22.91	450.00	10.31
B	15.00	-0.50	26.63	-6.53	450.00	-2.94
C	15.00	-0.70	26.63	-11.05	450.00	-4.97
D	15.00	-0.70	26.63	-11.05	450.00	-4.97

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_0} = b \times H = 450.00 \text{ ft}^2$$

Projected vertical area of roof

$$A_{vert_r_0} = 0.00 \text{ ft}^2$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 7.20 \text{ kips}$$

Leeward net force

$$F_l = F_{w,wB} = -2.9 \text{ kips}$$

Windward net force

$$F_w = F_{w,wA} = 10.3 \text{ kips}$$

Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 13.2 \text{ kips}$$

Roof load case 3 - Wind 90, GC_{pi} 0.18, $-C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (-ve)	15.00	-0.90	26.63	-25.17	225.00	-5.66
B (-ve)	15.00	-0.90	26.63	-25.17	225.00	-5.66
C (-ve)	15.00	-0.50	26.63	-16.11	450.00	-7.25

Total vertical net force

$$F_{w,v} = -18.58 \text{ kips}$$

Total horizontal net force

$$F_{w,h} = 0.00 \text{ kips}$$

Walls load case 3 - Wind 90, GC_{pi} 0.18, $-C_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient C_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A	15.00	0.80	26.63	13.32	450.00	5.99
B	15.00	-0.50	26.63	-16.11	450.00	-7.25
C	15.00	-0.70	26.63	-20.64	450.00	-9.29
D	15.00	-0.70	26.63	-20.64	450.00	-9.29

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_90} = d \times H = 450.00 \text{ ft}^2$$

Projected vertical area of roof

$$A_{vert_r_90} = 0.00 \text{ ft}^2$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 7.20 \text{ kips}$$

Leeward net force

$$F_l = F_{w,wB} = -7.3 \text{ kips}$$

Windward net force

$$F_w = F_{w,wA} = 6.0 \text{ kips}$$

Overall horizontal loading

$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 13.2 \text{ kips}$$

Roof load case 4 - Wind 90, $GC_{pi} -0.18, +c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A (+ve)	15.00	-0.18	26.63	0.72	225.00	0.16
B (+ve)	15.00	-0.18	26.63	0.72	225.00	0.16
C (+ve)	15.00	-0.18	26.63	0.72	450.00	0.32

Total vertical net force $F_{w,v} = 0.65$ kips

Total horizontal net force $F_{w,h} = 0.00$ kips

Walls load case 4 - Wind 90, $GC_{pi} -0.18, +c_{pe}$

Zone	Ref. height (ft)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (psf)	Net pressure p (psf)	Area A_{ref} (ft ²)	Net force F_w (kips)
A	15.00	0.80	26.63	22.91	450.00	10.31
B	15.00	-0.50	26.63	-6.53	450.00	-2.94
C	15.00	-0.70	26.63	-11.05	450.00	-4.97
D	15.00	-0.70	26.63	-11.05	450.00	-4.97

Overall loading

Projected vertical plan area of wall

$$A_{vert_w_90} = d \times H = 450.00 \text{ ft}^2$$

Projected vertical area of roof

$$A_{vert_r_90} = 0.00 \text{ ft}^2$$

Minimum overall horizontal loading

$$F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 7.20 \text{ kips}$$

Leeward net force

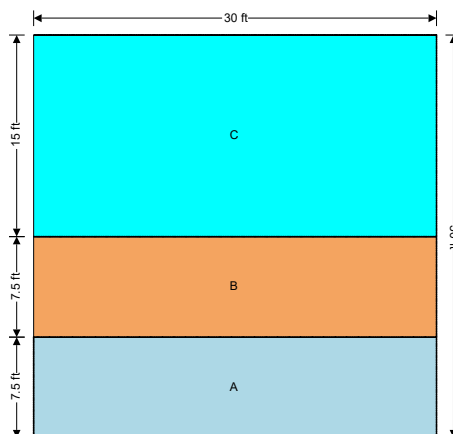
$$F_l = F_{w,wB} = -2.9 \text{ kips}$$

Windward net force

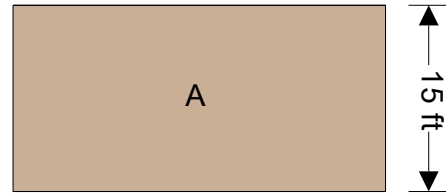
$$F_w = F_{w,wA} = 10.3 \text{ kips}$$

Overall horizontal loading

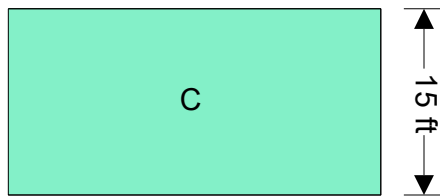
$$F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 13.2 \text{ kips}$$



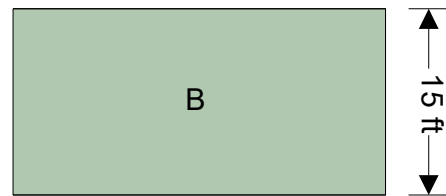
Plan view - Flat roof



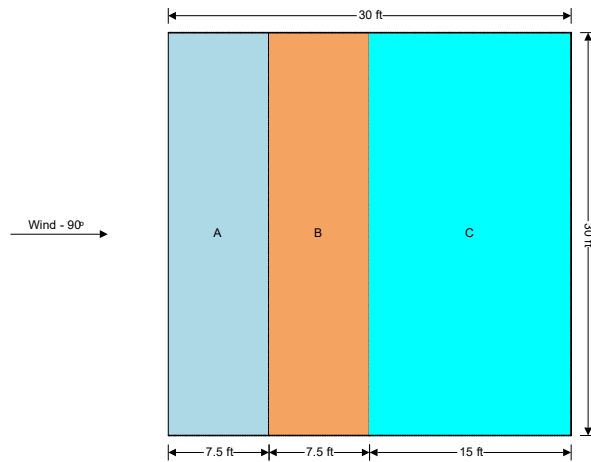
Windward face



Side face



Leeward face



Plan view - Flat roof

